Experiment 5: Transistor Small-Signal Amplifiers

Malachy Crossan & Brandon Saavedra

EEE3307-C0013: Electronics I

Due Date: 2026-11-10
Last modified: 2026-11-10

5.A Objective

5.B Pre-Lab

5.B.1 Common-Emitter Amplifier

5.B.1.a Calculate R_1 and R_2

Targeting $I_{\text{CQ}} \approx 1mA$

$$\begin{aligned} R_{\text{TH}} &= .1 \times (1+\beta) \times R_E \\ &= .1 \times 151 \times 1.8k\Omega \approx 27.2k\Omega \end{aligned}$$

Eq 1

$$\begin{aligned} I_B &= \frac{1mA}{150} = 6.67\mu A \\ I_E &= I_B \times (1+\beta) = 1.007mA \end{aligned}$$

Eq 2

$$\begin{aligned} V_{\text{TH}} &= V_{\text{BE}} + I_E \times R_E + I_B \times R_{\text{TH}} \\ &= .7 + 1.81 + .18 = 2.69V \end{aligned}$$

Eq 3

$$\begin{aligned} V_{\text{TH}} &= V_{\text{CC}} \times \frac{R_2}{R_1 + R_2} \\ R_1 &= 121.249k\Omega \approx 120k\Omega \\ R_2 &= 35.033k\Omega \approx 36k\Omega \end{aligned}$$

Eq 4

5.B.1.b Compute Q-point and max unclipped output voltage

5.B.1.c Compute the small-signal voltage gain $A_{\text{vi}} = \frac{v_o}{v_i}$ and $A_{\text{vs}} = \frac{v_o}{v_s}$

5.B.1.d Compute the input and output resistances

5.B.2 Common-Emitter Amplifier without C_E

5.B.2.a Compute Q-point and max unclipped output voltage

5.B.2.b Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

5.B.2.c Compute the input and output resistances

5.B.3 Common-Collector (Emitter Follower) Amplifier

5.B.3.a Calculate R_1 and R_2

5.B.3.b Compute Q-point and max unclipped output voltage

5.B.3.c Compute the small-signal voltage gain Avi = vo/vi and Avs = vo/vs

5.B.3.d Compute the input and output resistances

5.C Simulation

5.C.1 Common-Emitter Amplifier

5.C.2 Common-Emitter Amplifier without C_E

5.C.3 Common-Collector (Emitter Follower) Amplifier

5.D Experimental Procedure